KOWSHIN V

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CAREER OBJECTIVE

Aspiring VLSI and Embedded Systems Engineer with hands-on experience in RTL Design, ASIC Design Flow, and Embedded Programming. Seeking opportunities to develop innovative semiconductor and embedded solutions.

EDUCATION QUALIFICATION

BE ECE - Hindusthan College of Engineering and Technology, Coimbatore — 2023 - 2027
CGPA: 8.30/10 (Up to VI Semester)

HSE - Saraswathi Matric Hr. Sec. School, Mayiladudhurai — 2021 - 2023
Percentage: 81.5%

SSLC - Saraswathi Matric Hr. Sec. School, Mayiladudhurai — 2020 - 2021
Percentage: Pass

TECHNICAL SKILLS

VLSI DESIGN

- Verilog HDL (RTL Design)
- Digital Design
- ASIC Design Flow (RTL-to-GDSII)
- RTL Verification (Testbench & Waveform Analysis)
- SpyGlass (Lint, CDC & RDC)
- RTL Synthesis & Physical Design
- Timing Analysis: LEC & GLS

EDA Tools: Synopsys (SpyGlass, VCS, Design Compiler), Cadence (Genus, Innovus, SimVision, Conformal), Xilinx Vivado

EMBEDDED SYSTEMS AND IOT

- Embedded C Programming
- ESP32 & Arduino Programming
- Sensor Interfacing
- GPIO, UART, I2C & SPI Communication
- IoT Fundamentals

Programming Languages: C programming, Embedded C, Python (Basic)

INTERNSHIP EXPERIENCE

ANGSTROMERS TECHNOLOGIES BANGALORE - VLSI DESIGN ENGINEER — Jul 2026 – Present

- Designed and verified an Approximate Tensor Processing Unit (APTPU) using Verilog HDL.
- Performed RTL verification, Lint, CDC & RDC analysis using Cadence SimVision and Synopsys SpyGlass.
- Executed RTL synthesis using Cadence Genus and Physical Design using Cadence Innovus.
- Performed LEC, Gate-Level Simulation (GLS) and timing verification for ASIC signoff.

NLC INDIA LIMITED-ELECTRONICS & CONTROL DEPARTMENT — Aug-2025

- Studied industrial instrumentation and control systems used in thermal power plants.
- Observed SCADA and DCS-based automation systems.
- Gained practical exposure to industrial electronic control and monitoring systems.

CERTIFICATIONS (NPTEL ELITE - ONLINE) Jul 2024 - Oct 2025

- **Fundamental of AI**: AI basics, search algorithms, reasoning techniques, and introductory machine learning.
- **Cloud Computing**: Gain Storage, virtualization, and cloud-based IoT application deployment.
- **IOT**: Sensor integration, device communication, and real-time data monitoring.

PROJECTS

Approximate Tensor Processing Unit (APTPU) – RTL-to-GDSII Tapeout

- Designed and implemented a 12×12 Approximate Processing Element (APE) Systolic Array using Verilog HDL.
- Successfully completed the ASIC RTL-to-GDSII flow, including RTL Verification, SpyGlass (Lint, CDC & RDC), RTL Synthesis, Physical Design, LEC, GLS, and a timing-clean, tapeout-ready implementation..

UART Transmitter and Receiver using Verilog HDL

- Designed and implemented UART Transmitter and Receiver using Verilog HDL with FSM-based RTL architecture.
- Performed functional simulation and testbench verification using Synopsys and Cadence tools.

Smart Geo-Fence Alert for Fisherman

- Developed a GPS-based geo-fencing system to detect maritime boundary crossings and enhance fisherman safety.
- Integrated GSM alerts and buzzer notifications for real-time emergency communication.

ACHIEVEMENTS

Winner of the IVTon – Urbanisation & Smart Cities Track at graVITas 2025, VIT Vellore, for developing an innovative smart city solution. This achievement demonstrates creativity, technical excellence, problem-solving ability, and effective teamwork in addressing real-world urban challenges.

SOFT SKILLS

Problem Solving, Analytical Thinking, Teamwork, Communication Skills, Time Management, Adaptability, Quick Learning

PERSONAL DETAILS

- **Date of Birth:** 01 October 2005
- **Languages Known:** Tamil (Native), English (Professional)
- **Address:** Mayiladuthurai, Tamilnadu

DECLARATION

I hereby declare that the information furnished above is true and correct to the best of my knowledge and belief.

Place: Mayiladuthurai
Date: _____

(Kowshin V)